

Whether treatment holidays help depends on three measurable properties of resistant cells: a single-cell simulation of two clinical dosing trials

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Abstract

A melanoma trial and a set of melanoma xenografts gave opposite verdicts on intermittent dosing, and a glioblastoma trial found no benefit from a dose-dense temozolomide regimen designed to deplete MGMT. We ask what a mechanistic single-cell model has to assume to produce each of those outcomes. Resistance here is a heritable expression state with single-cell kinetics rather than a compartment with an assumed rate: a slowly switching promoter whose protein sets a cell's death hazard, inherited through division in cells that grow, replicate their genes and divide. Whether treatment holidays help is decided by three properties of resistant cells — their growth under drug, their fitness without it, and whether the drug itself stabilises the resistant state — and setting those three reproduces both the clinical and the preclinical melanoma outcome, the preclinical one as a lower end-of-simulation burden rather than as a longer time to loss of control. In glioblastoma, modelling MGMT as the suicide enzyme it is creates a dose-dense advantage in unmethylated tumours that the trial did not find, which bounds the per-lesion rate at which temozolomide consumes MGMT; in methylated tumours the model already diverges from the trial with MGMT stable, on cumulative dose alone, so the null there bounds the model's dose response instead. The mechanisms are set in each case rather than inferred from the outcomes, so these are demonstrations that the rankings follow from cell properties, not claims that particular tumours carry them. What the framework adds is that those properties are measurable in lineage-tracing and growth experiments, and can therefore be bounded before a schedule is chosen.

Author summary

Should a cancer drug be given continuously, or with planned breaks? Trials disagree. In melanoma, a randomised trial found continuous dosing better, while mouse experiments had pointed the other way. In glioblastoma, a regimen giving twice as much drug, to exhaust a repair protein that protects cells, did no better than the standard one.

We asked what a model of individual cells has to assume to produce each result. Here a cell is resistant because of which genes it happens to be expressing, a state it passes to its daughters, rather than because it belongs to an abstract “resistant compartment”. That makes the assumptions concrete enough to measure.

Three properties decide whether breaks help: how fast resistant cells grow under the drug, how well they do without it, and whether the drug locks the resistant state in place. Set those three, and the model gives either answer, which is why the trials disagreeing is not a paradox. For glioblastoma, the trial's negative result limits how fast the drug uses up the repair protein, but only where that protein is common; where it is rare the model disagrees with the trial for a different reason, and we say so.

Cancer cells that survive targeted or cytotoxic therapy frequently do so without a resistance mutation: they occupy a rare, reversible, drug-tolerant expression state that pre-exists the treatment or is induced by it [1–5]. Because such a state is reversible, when the drug is given matters as much as how much of it is given, and two lines of evidence on that question point in opposite directions. In melanoma, intermittent BRAF-inhibitor dosing prevented the emergence of resistance in patient-derived xenografts, in which established resistant tumours regressed when the drug was withdrawn [6]; in the randomised SWOG S1320 trial, continuous dosing gave the longer progression-free survival [7]. In glioblastoma, the RTOG 0525 trial compared the standard temozolomide regimen with a dose-dense one designed to deplete the repair protein MGMT, delivering about twice the cumulative dose, and found no survival difference in either methylated or unmethylated tumours [8].

Compartment models of adaptive therapy established the principle that governs the first of these: holidays pay when resistance carries a fitness cost, and tumour turnover sets how large that cost has to be [9,10]. Phenotypic-switching theory added that perturbing how long a resistant state is retained can itself decide an outcome [11]. What those treatments leave abstract is the resistant compartment: its size and its switching rates are fitted quantities of the model rather than properties of cells that an experiment measures. Lineage-resolved experiments now measure exactly those properties — the correlation of fates between sisters and cousins, the collapse or preservation of clonal barcode diversity, and the timescale of expression fluctuations relative to the cell cycle [2,4,12,13] — so a model whose resistant state is a promoter rather than a compartment can be confronted with them.

Here we put the two dosing questions to such a model. Resistance is one slowly switching gene whose protein sets a cell’s death hazard, in cells that grow, replicate their genes, divide and partition their molecules, with a dose-schedule layer and a schedule optimiser on top; the framework, its validation against exact stationary laws, and the fit of the resistance-gene model to time-lapse measurements of cisplatin-treated cells are reported in a companion Article. We do not fit these trials. We ask instead what the model has to assume to produce each reported outcome, and what the outcomes therefore bound. For melanoma the answer is three properties of resistant cells, all of them measurable; for glioblastoma it is a molecular rate constant that has not been measured directly, and a place where the model and the trial disagree for a reason that has nothing to do with MGMT.

Results

Treatment holidays help or hurt according to three cell properties

The forward model used here is the one described in the companion Article: stochastic reaction kinetics inside cells that grow, replicate their genes and divide, with a single resistance gene whose promoter switches slowly, whose protein protects the cell through a concentration-dependent death hazard, and whose state both daughters inherit at division. On top of it sits a dose-schedule layer and a schedule optimiser (Methods). What that combination buys is the subject of this Article: a resistance phenotype with measurable single-cell kinetics rather than a compartment with an assumed rate, so that the properties which decide a dosing question become quantities a lineage experiment can bound. We first asked which schedule best delays regrowth of a melanoma-like population with a rare, heritable pre-resistant state [2] treated at a fractional-killing dose, comparing continuous dosing, the intermittent schedule of the SWOG S1320 trial, in which patients received a continuous 8-week lead-in and were then randomised to continuous dosing or to cycles of 3 weeks off and 5 weeks on [7], and adaptive therapy, in which the drug is withdrawn when the population has halved and resumed when it regrows to its initial size [9]. Because adaptive therapy deliberately lets the tumour regrow, schedules are compared by the time after randomisation at which the population exceeds 120% of its pre-treatment size (loss of control), with the RECIST-like time to progression from the nadir reported alongside (Methods).

The answer depends on three properties of the resistant cells (Fig. 1a–d): their growth under drug, their fitness without it, and whether the drug stabilises the resistant state. When the resistant state is fully protective and carries no fitness cost, as lineage data indicate for pre-existing tolerance [12], resistant cells grow at the same rate with or without drug, so a holiday cannot cost the resistant subpopulation anything and can only let sensitive cells regrow to be killed again. Even so, both interrupted schedules keep control longer than continuous dosing: the three lose control after 23, 26 and 27 weeks, with the intermittent schedule delivering 65% of the cumulative dose of continuous dosing over the full 60-week simulation. The reason is a second drug effect rather than the fitness of resistant cells: the drug stabilises the resistant state tenfold by suppressing the promoter inactivation rate (Methods), so continuous exposure keeps cells resistant that a holiday allows to revert. That effect is set in the model, not fitted, and it is what makes holidays worth anything at all in this arm. When resistant cells pay a 50% fitness cost in the absence of drug, a coarse stand-in for the drug dependence that made intermittent dosing superior in vemurafenib-treated patient-derived xenografts, where withdrawal drove established resistant tumours to regress rather than merely grow more slowly [6], holidays let sensitive cells outcompete them. On the declared end point the three schedules are indistinguishable: none loses control within the 52-week follow-up (52+, 52+ and 52+ weeks, censored). The xenograft ranking appears instead in the tumour burden at the end of the 60-week simulation, which the intermittent schedule leaves at 0.35 of the pre-treatment size against 0.77 under continuous dosing, on 65% of its cumulative dose; adaptive therapy, which withdraws the drug earliest, ends higher at 0.83 but on 36% of the cumulative dose of continuous dosing. When resistant cells are only partly protected and grow at half speed under drug, holidays release them: continuous dosing keeps control for 52+ weeks against 49 for the intermittent schedule and 43 for adaptive therapy, the ranking observed in the trial, where continuous dosing gave a median progression-free survival of 9.0 months against 5.5 months for intermittent dosing. Scored from the nadir instead, as in RECIST, the same simulations give 9, 11 and 4 weeks for continuous, intermittent and adaptive dosing without a fitness cost, 16, 26 and 4 weeks with a fitness cost, and 15, 12 and 4 weeks under partial protection. Adaptive therapy progresses from its nadir almost at once in every mechanism, because the schedule withholds the drug until the population has regrown to its pre-treatment size, and in the fitness-cost arm all three schedules progress from the nadir although none loses control within the follow-up. That discordance is why the schedules are compared by loss of control: an end point referred to the nadir penalises a schedule for the regrowth it is designed to permit. Gating three quarters of the death hazard to a window of the cell cycle, as in the persister simulations of the companion Article, leaves these comparisons standing, whether or not the maximal hazard is raised to hold the cycle-averaged hazard fixed (Supplementary Note 1 and Supplementary Fig. 1): the best schedule is unchanged in 3 of 3 mechanisms and the model still favours continuous dosing under partial protection (52 against 43 weeks, from 52 and 49). The memory of the resistant state, five net population doublings, is an assumption rather than a measurement, and it is what decides whether resistant cells revert during a three-week holiday; scanning it from two to twelve doublings leaves the ranking unchanged in the two arms scanned, without and with a fitness cost of resistance, and narrows the advantage of holidays from 9 weeks at 2 net doublings of memory to 1.7 weeks at 12; the partial-protection arm, the one that reproduces the ranking of the trial, was not scanned (Supplementary Note 2). That the benefit of holidays should depend on a cost of resistance is established by compartment models of adaptive therapy, which also show that tumour turnover modulates how large that cost must be [10]. Here the dependence emerges from single cells whose resistance is a heritable expression state rather than a fixed compartment, the cost and the degree of protection are properties of cells that lineage experiments measure, and the third case shows that partial protection reverses the ranking even when a cost is absent. The optimiser makes the dependence explicit (Fig. 1e): over intermittent schedules of any period and duty cycle after the lead-in, the best schedule without a fitness cost is a period of 2.2 weeks with 39% of the time on drug (27 weeks to loss of control). With a fitness cost, no best schedule is identified: 12 of the 16

grid points reach the 52-week censoring limit without loss of control, at every duty cycle of 47% or more and at every period from 1.0 to 12 weeks; with partial protection, continuous dosing is the only schedule that keeps control for the whole 52-week follow-up: the 4 of the 16 grid points that reach the censoring limit are all at a duty cycle of 100%, which is the same schedule whatever the nominal period. With a fitness cost the star in Fig. 1e is therefore the first point of a tie and not an optimum, and what the scan establishes there is that control is kept for the whole follow-up, not which schedule keeps it best; under partial protection the tie is between four grid points that are all continuous dosing, so there the star does mark the winning schedule. A bacterial-persistence model adapted to colorectal-cancer lines reaches the same conclusion by another route: an intermediate dose with release periods is optimal, because the rate of transition into the persistent state rises with concentration and then saturates [14]. The agreement is partial and the difference is informative. Where tolerance is drug-induced, the optimiser here also selects an intermediate dose over the highest one, for the same reason. Release periods, however, help there only when the drug changes the kinetics of the resistant state, either by stabilising it, as in the melanoma arm above, or by inducing it at a rate that saturates, as in that model; with a state whose decay rate is the same with and without drug, a holiday merely postpones the death of reverting cells while every cell grows, and the optimiser rejects it. The model therefore reproduces both the clinical and the preclinical outcome, the preclinical one as a lower burden at the end of the simulation rather than as a longer time to loss of control, and shows their difference to depend on measurable properties of the resistant cells, their growth under drug, their fitness without it and whether the drug stabilises their state, properties of cells rather than of the tumour, which growth and lineage experiments can bound, within the identifiability limits reported in the companion Article, before a schedule is chosen. The mechanism is set in each case, not inferred from the outcome, so this is a demonstration that the ranking follows from cell properties rather than a claim that these particular cells carry them; and because the trial and the xenograft also differ in drug and species (dabrafenib plus trametinib in patients, vemurafenib monotherapy in mice), their contrast is not attributable to resistant-cell fitness alone.

Consumption of MGMT, not dose, is what the glioblastoma trial bounds

The second case study concerns glioblastoma, where the RTOG 0525 trial compared the standard temozolomide regimen (150–200 mg/m² on days 1–5 of 28-day cycles) with a dose-dense regimen (75–100 mg/m² on days 1–21) that delivers about twice the cumulative dose and was designed to deplete MGMT, and found no difference in survival for either MGMT-methylated or unmethylated tumours [8]. We simulated both regimens with one-compartment pharmacokinetics (elimination half-life 2.1 h, a plasma value from a three-compartment population model of plasma and cerebrospinal fluid [15]; an independent single-dose study gives 1.8 h [16]) on populations in which 1% (methylated) or 30% (unmethylated) of cells express MGMT. Those two fractions are illustrative values that bracket low and high expression, not measurements, and the idealisation matters: promoter methylation shifts MGMT expression as a graded quantity across cells rather than switching a discrete subpopulation on, so the whole dose-dense prediction below rests on treating it as two states. With that caveat, with MGMT protein protecting cells from the drug [17] and either stable or consumed by it stoichiometrically, one molecule per lesion repaired, as a suicide enzyme is, so that the pool follows the cumulative exposure rather than the peak concentration (Fig. 2a,b). Lesions are not tracked as a state variable: they are taken to form in proportion to dose and to be repaired at once, so “one molecule per lesion” enters only through the rate of consumption, and no unrepaired damage accumulates. The rate constant of that consumption is not fitted to data; it is set so that consumption at the standard bolus peak matches the first-order parameterisation it replaces, which makes the two mechanisms comparable at the same initial rate and isolates how each integrates over time. With stable MGMT, the depth of response over six cycles is 0.20 log₁₀ units for the standard regimen and 1.00 for the dose-dense regimen in the methylated population (0.08 and 0.10 in the unmethylated one), and a 21-day regimen at the

165 standard cumulative dose achieves no measurable kill ($0.00 \log_{10}$ units): the death hazard rises steeply with
 166 concentration, so spreading the same dose over more days loses efficacy, and the advantage of the dose-dense
 167 arm in the model is entirely its larger cumulative dose. With stable MGMT the two arms therefore differ little
 168 in the unmethylated population (0.08 against $0.10 \log_{10}$ units), as in the trial. In the methylated population,
 169 where almost no cell is protected, they differ a great deal. The death hazard is already close to saturation at
 170 both daily peaks, so halving the dose per day costs less than quadrupling the number of dosing days gains, and
 171 the dose-dense arm reaches 1.00 against $0.20 \log_{10}$ units on cumulative dose alone, with MGMT stable and no
 172 consumption at all. RTOG 0525 found no advantage in the methylated subgroup either, so that discrepancy
 173 belongs to the model’s mapping from dose to kill, and to the exposure it assumes was delivered, rather than to
 174 anything about MGMT. Consumption changes that. Because one molecule is spent per lesion repaired, the pool
 175 follows the cumulative exposure, which the dose-dense regimen more than doubles: in the methylated population
 176 its response deepens to $1.52 \log_{10}$ units against 0.22 for the standard regimen, and in the unmethylated population,
 177 where 30% of cells start protected, the same depletion turns a regimen that barely worked into one that controls
 178 the population (1.08 against $0.18 \log_{10}$ units; final size 0.1 against 1.1 times the initial size, and 4.9 and 3.6
 179 with stable MGMT). Modelled as the suicide enzyme it is, MGMT therefore creates a dose-dense advantage
 180 where the model had none: in the unmethylated population, consumption alone separates two regimens that the
 181 stable model, like the trial, ranked as equivalent. That is the inference the trial can carry, and read as a bound
 182 rather than as a fit it is informative: either temozolomide consumes MGMT in unmethylated tumours far more
 183 slowly than this stoichiometry allows, or the benefit of depleting it is offset by something the model leaves out.
 184 It cannot be carried over to the methylated population, where the divergence from the trial is already present
 185 with MGMT stable and is merely deepened by consumption (1.52 against $1.00 \log_{10}$ units for the dose-dense
 186 arm), so the null there bounds the consumption rate not at all. At the standard cumulative dose the number of
 187 dosing days per cycle that minimises the final population is 9 (methylated) and 9 (unmethylated) with stable
 188 MGMT, and 9 and 5 when MGMT is consumed (Fig. 2c). These simulations are not fits to the trial, whose end
 189 points depend on much that the model omits, dose-limiting toxicity and the dose reductions it forces among
 190 it: grade 3 or worse events were far commoner in the dose-dense arm, 53% against 34%, mostly lymphopenia
 191 and fatigue [8], while both arms here receive their nominal dose for all six cycles, so the twofold difference in
 192 exposure that drives the modelled separation need not be the difference the patients received. What they add
 193 is a sharper version of the rationale for dose-dense dosing, and it holds for unmethylated tumours alone: there
 194 the benefit that dose-dense dosing is supposed to deliver exists in the model only at consumption rates that the
 195 trial’s null result appears to exclude. In methylated tumours the modelled advantage comes from cumulative dose
 196 rather than from depletion, so its absence in the trial bounds nothing about MGMT and marks instead the limit
 197 of a dose response calibrated on the peak concentration. The quantity the argument rests on is the per-lesion
 198 molecular rate constant of that consumption, which has not been measured directly, although bulk depletion
 199 time-courses have: protracted temozolomide lowers alkyltransferase activity in peripheral blood mononuclear
 200 cells by about 63% at 14 days and 73% at 21 days [18], a compressed $1,000 \text{ mg/m}^2$ schedule takes it to 18% of
 201 initial levels while O^6 -methylguanine accumulates in both blood cells and tumour biopsies [19], and in orthotopic
 202 GBM43 xenografts tumour MGMT activity is depleted by day 6 on both the standard and the 21-day schedule
 203 and stays suppressed only on the 21-day one [20]. The last of these is the sharpest comparison available, and it
 204 makes the disagreement more specific than the trial alone does: the deeper and longer tumour depletion that the
 205 dose-dense schedule is supposed to deliver was confirmed in that model, and survival still did not improve. Either
 206 the depletion measured in bulk is not the per-cell depletion that matters for killing, or the benefit of depleting
 207 MGMT is offset by something the model leaves out.

Discussion

The same dosing question has opposite answers in these simulations depending on three properties of the resistant cells, and a trial result and a xenograft result that appeared to contradict each other are both reproduced once those properties are set, the xenograft result as a lower burden at the end of the simulation rather than as a longer time to loss of control. The properties are set in each case rather than inferred from the outcomes, so what the case studies show is that the ranking follows from cell properties, not that any particular tumour has them. Neither anchor is as one-sided as a single end point makes it look. In SWOG S1320 the progression-free survival advantage of continuous dosing (median 9.0 against 5.5 months, hazard ratio 1.36 for intermittent against continuous, $P = 0.063$ against the trial’s pre-specified two-sided α of 0.2) did not carry through to overall survival, a secondary end point for which the trial was not powered and on which the two arms were equal at a median of 29.2 months [7]; and in the xenografts the resistant tumours were drug-addicted, regressing on withdrawal rather than merely growing more slowly [6], which is the extreme of the fitness cost that the second mechanism here represents. The two studies also differ in drug and species, so their contrast is not attributable to resistant-cell fitness alone.

What a single-cell forward model adds to the compartment models that established the principle is that the deciding quantities stop being parameters of the model and become properties of cells. Growth under drug, fitness without it and drug-induced stabilisation of the resistant state are all measurable in the growth and lineage-tracing experiments that are now routine, so a schedule question can be narrowed before a trial is designed rather than settled after one. That is the practical claim of this Article, and it is bounded in an important way: the companion Article shows that fitting this model to fate fractions alone leaves its memory timescale undetermined and admits a cycle-gated hazard as an equally good explanation. The properties that decide a schedule are measurable, but they have not been measured here, and the optimiser is only worth running on parameters an experiment has pinned down.

The glioblastoma study is the sharper case, because the model and the trial disagree in two different ways and only one of them is informative. Modelled as the suicide enzyme it is, MGMT creates a dose-dense advantage in unmethylated tumours where the stable model, like the trial, ranked the two regimens as equivalent, so the trial’s null bounds the per-lesion rate of consumption there. In methylated tumours the model separates the regimens on cumulative dose alone, with MGMT stable, and the trial found no advantage there either; that discrepancy constrains the model’s dose response and the exposure it assumes was delivered, not the rate of consumption. Reporting the second as though it were the first would turn a limitation into a result. The clinical case studies use literature values for cell-cycle times, resistant fractions and pharmacokinetics rather than patient-specific measurements, so their conclusions concern mechanisms and rankings, not absolute survival times. In the glioblastoma study MGMT is consumed stoichiometrically, one molecule per lesion repaired, so its depletion follows the cumulative exposure as a suicide enzyme’s should. Two things about that model should be read with care. Lesions are not a state variable: they are assumed to form in proportion to dose and to be repaired immediately, so “one molecule per lesion” enters only through the consumption rate and no unrepaired damage accumulates. And the rate constant of that consumption was set so that the initial consumption rate at the standard bolus peak matches the first-order parameterisation it replaces, which makes the two mechanisms comparable but leaves the constant itself unmeasured; bulk depletion time-courses in blood cells and in tumour tissue exist [18–20] and bound it loosely, and that is why we name the per-lesion constant as the quantity that separates the regimens in unmethylated tumours. In methylated tumours the model separates them even with MGMT stable, on cumulative dose alone, and RTOG 0525 found no advantage there either; that discrepancy constrains the model’s dose response and the exposure it assumes was delivered, not the rate of consumption.

MGMT is in any case only one determinant of a temozolomide response that is also mismatch-repair-dependent, and the 1% and 30% expressing fractions are illustrative values that bracket low and high expression rather than measured quantities: promoter methylation shifts expression as a graded quantity across cells rather than switching a discrete subpopulation on, and the dose-dense prediction rests entirely on this two-state idealisation. The memory of the resistant state is expressed in these tumour-scale studies as a number of net population doublings, not of cell divisions, and so is not the division-counted memory of memory-sequencing experiments. Because the framework simulates the lineage-resolved data from which the deciding properties are measured, the natural next step is not a larger scan but an experiment: a lineage-tracing measurement in which the growth of resistant cells with and without drug, and the stability of the resistant state under exposure, are read off the same tree. Those three numbers are what stand between a schedule hypothesis and a schedule prediction.

Methods

The forward model, its kernels and the fit of the resistance-gene model to time-lapse data are described in full in the companion Article; the sections below give what the schedule studies of this Article add to it, and repeat the model layers they depend on.

Reaction model and propensities

A model is a set of species and reaction channels. Species are molecules (partitioned at division) or promoter states (inherited). For a reaction with reactant counts x_i , stoichiometric coefficients ν_i , rate constant k and volume exponent e , the mass-action propensity is

$$a = k V^e \prod_i \binom{x_i}{\nu_i},$$

where by default $e = 1 - \sum_i \nu_i$, so that zero-order production scales with volume, first-order reactions are volume independent and bimolecular reactions scale as $1/V$; this keeps concentrations invariant under growth. Regulated reactions multiply this by $b + (1 - b)F(\mathbf{c})$, where b is a basal fraction and F combines Hill terms of the regulator concentrations $c_j = x_j/V$, $h_j = c_j^{n_j}/(K_j^{n_j} + c_j^{n_j})$ for activators and $1 - h_j$ for inhibitors, either as a product (AND logic) or as $1 - \prod_j (1 - h_j)$ (OR logic). Promoter switching is represented by explicit promoter species (for instance $G_{\text{off}} \rightleftharpoons G_{\text{on}}$) grouped into promoter groups. Custom propensities can be supplied as functions of counts, volume, parameters and time.

Stochastic simulation kernels

Within one update step the cell volume and gene copy number are held fixed and the reaction network is advanced by one of four kernels: the direct method [21] with propensities updated through a species-reaction dependency graph; fixed-step Poisson tau-leaping [22]; a hybrid scheme in which a leap that would drive any species negative is discarded and that interval is simulated exactly instead (the “switching” scheme of version 1, with the discarded interval resolved exactly rather than clipped); and adaptive tau-leaping with the step-size selection of [23] and the critical-reaction partition of [24], falling back to a fixed number of exact events when the proposed leap is shorter than $10/a_0$. The hybrid scheme is approximate rather than exact: a leap is kept only when it leaves every species non-negative, so an accepted step follows the Poisson leap truncated to the non-negative region, and only the discarded intervals follow the exact process. The discrepancy falls with τ and is bounded by the Kolmogorov-Smirnov test of Fig. 2d, which the scheme passes at $\tau = 0.05$ and fails at $\tau = 0.2$ for the models

tested; the direct method is used wherever exactness is required (Supplementary Table 1 lists the kernel correctness tests). Every kernel takes an explicit random number generator.

Cells, growth, replication and division

Each cell carries its volume V , age, generation, lineage identifiers, gene copy state, growth rate, molecule counts, a cell-specific parameter vector and its own random number generator (seeded from the parent’s generator, which makes results independent of the number of threads). Volume grows exponentially, $\dot{V} = \lambda V$, with an optional log-normal spread of λ between cells. Division is triggered by a sizer ($V \geq V_{\text{div}}$), an adder ($V - V_{\text{birth}} \geq \Delta$) or a timer ($\text{age} \geq T$), each with a per-cell noise factor drawn at birth. At division the first daughter inherits a volume fraction $f \sim \mathcal{N}(0.5, \sigma^2)$ (clipped to $[0.05, 0.95]$) and each molecule is assigned to it independently with probability f (binomial partitioning); a beta-binomial option models over-dispersed partitioning. Promoter groups are inherited by both daughters when unreplicated and split one copy per daughter, in random assignment, when replicated. Gene replication, if enabled, doubles the promoter counts (and the propensity of reactions flagged as copy-number dependent) once the cell has completed a set fraction of its cycle, and is reset at division. Populations can be started from per-founder states and volumes (for instance a snapshot of a burnt-in constant-size population). Population control is either constant size (every second daughter replaces a uniformly chosen cell, the behaviour of version 1), free growth (all daughters kept, with uniform subsampling above a cap and book-keeping of the true size), logistic (free growth with an additional hazard $\lambda N/K$) or a mother machine (one daughter kept at random at every division, so that each founder gives one statistically independent lineage and the recorded cells follow the single-lineage rather than the population snapshot distribution).

Drug and perturbation layer

A perturbation combines a dose schedule $d(t)$ (constant, pulsed, piecewise, one-compartment bolus pharmacokinetics or an arbitrary function) with effects evaluated for every cell at every step: a death hazard

$$h = h_{\text{max}} \frac{d^m}{\text{EC}_{50}^m + d^m} \cdot \frac{K^q}{K^q + c_P^q},$$

where c_P is the concentration of a protective species (survival probability over a step is $e^{-h \Delta t}$); growth inhibition by $1/(1 + (d/\text{IC}_{50})^{m_g})$, whose Hill coefficient m_g is separate from the death coefficient m and is 2 in every simulation reported here; a dose-independent fitness cost of a cell state, $1 - c_{\text{max}} c_P^q/(K^q + c_P^q)$, on the growth rate; multiplicative modulation of any parameter by a function of the dose; and genetic perturbations (knockdown, knockout, over-expression) of any parameter in a random fraction of cells, the flag being inherited by daughters.

Two further effects couple the drug to the cell cycle and to the protein that resists it. A cell-cycle-dependent hazard multiplies h by

$$\beta + (1 - \beta) \exp \left[-\frac{1}{2} \left(\frac{\varphi - \varphi_0}{w} \right)^2 \right],$$

where φ is the fraction of the cell’s volume increment completed, $(V - V_{\text{birth}})/(V_{\text{div}} - V_{\text{birth}})$ under a sizer and $(V - V_{\text{birth}})/\Delta$ under an adder, and the fraction of the interdivision time elapsed under a timer (0 at birth, 1 at division), so that killing peaks at φ_0 and falls to a fraction β of the peak elsewhere; $\beta = 1$ is the cycle-independent hazard and $\beta = 0$ confines killing to a window of width w around φ_0 . This represents an agent whose lesions are converted into death during replication. Stoichiometric consumption of a protective protein removes molecules of it at rate

$$k d V \frac{c_P}{c_P + K_m},$$

by a Poisson draw over the step. Because lesions form in proportion to the dose, the rate is set by lesion formation alone while the protein is abundant, so the pool follows the cumulative exposure rather than the peak concentration, and falls with what remains once the pool runs low; resynthesis is whatever the reaction model provides. This is the behaviour of a suicide enzyme such as MGMT, which is inactivated by the very lesion it repairs.

Clinical regimens, treatment outcomes and schedule optimisation

Regimens are built from the schedule primitives of the perturbation layer: `PulsedDose` with a finite number of cycles, `PiecewiseDose` for arbitrary calendars (the S1320 schedule), `daily_boluses` and `cycle_days` for daily oral doses with one-compartment pharmacokinetics, and `AdaptiveDose`, a feedback schedule that withdraws the drug when the population falls below a fraction of its initial size and resumes it when the population regrows above another fraction, evaluated with the current population size at every step [9]. Outcomes computed from a simulation are the net growth rate, the depth of response ($\log_{10}$ of the ratio of the initial to the smallest population), the time to progression (first time the population exceeds a threshold multiple of its running minimum, the analogue of RECIST progression from the nadir; the multiple is an argument of the function, 1.2 by default, and the melanoma case study below sets it to 1.73; censored at the end of the simulation), the extinction probability over replicates and the cumulative dose actually applied. `optimize_schedule` minimises any such outcome over a box of schedule parameters: a log-spaced grid (or a Latin hypercube when the grid is large) is evaluated with the same random seeds for every candidate, and the best point is refined by Nelder-Mead in a bounded transform of the box; constraints such as a fixed cumulative dose enter as penalties or as the parameterisation (for instance a per-day dose equal to the budget divided by the number of dosing days). The case studies are run at the scale of a tumour rather than a culture: the cell cycle of the model is the net doubling time of the tumour (4 weeks for melanoma, 40 days for glioblastoma). The glioblastoma value lies within the volume doubling times measured for untreated tumours, a median of 21 days in one magnetic-resonance series and 50 days in another [25,26]; the melanoma value is an assumption, since net doubling times of treated cutaneous melanoma are not established, and it sets the time axis of the melanoma case study rather than the ranking of schedules, which follows from the resistance mechanism. Growth is exponential throughout, which is a simplification: the same measurements favour Gompertzian slowing in large glioblastomas [26], so absolute regrowth times in Figs 1 and 2 are shorter than a saturating model would give. Gene-expression rates are scaled to the hour (mRNA half-life 7 h and protein half-life 35 h, generic mammalian values; at the tumour time scale both species turn over many times per generation, so the results are insensitive to them and all heritable memory resides in the promoter), and founders are drawn from the stationary distribution of promoter states. The melanoma case uses a pre-resistant fraction of 1:200 (a per-marker frequency [2] used as a proxy for the functional pre-resistant fraction, which is not the same quantity, and 5- to 50-fold higher than the 1 in 1,000 to 1 in 10,000 at which drug-naive precursors of resistant fates have been traced [27]; the arms of Figs 1 and 2 that lose control therefore do so earlier than the traced frequency would give, while those censored at 52 weeks are unaffected), with a memory of five net doublings (the model's generation is the net doubling of the tumour, a modelling choice rather than the cell-division-counted memory of memory sequencing [28]) and drug effects at the fractional-killing dose: sensitive cells arrest and die with a half-life of two weeks, cells in the resistant state are protected, and the drug stabilises the resistant state ten-fold (the drug-induced reprogramming of [2]). Three variants of the resistant state are compared: full protection without cost; a 50% fitness cost in the absence of drug [6]; and partial protection, in which resistant cells grow at half speed under drug. Progression is scored in two ways: from the nadir, as in RECIST (a 20% increase in diameter, taken as a 1.73-fold increase in cell number, after the 8-week lead-in; a well-mixed cell count cannot represent RECIST's absolute 5 mm threshold, its new-lesion and non-target criteria or its sum-of-diameters basis,

and it assumes cell number proportional to volume, so this is an analogue rather than the clinical end point), and as loss of control, the time after randomisation at which the population exceeds 120% of its pre-treatment size, the end point of adaptive-therapy studies, which is the one used to compare schedules that deliberately let the tumour regrow. The two differ in their reference point as well as in their multiple: progression from the nadir is 1.73 times the running minimum of the population after randomisation, whereas loss of control is 1.2 times its size before treatment, which does not depend on how deep the response was. Both are computed for every arm of the melanoma study and both are reported. The temozolomide case uses a one-compartment elimination half-life of 2.1 h [15] (a plasma value from a three-compartment population model of plasma and cerebrospinal fluid; that study measured cerebrospinal fluid rather than tumour tissue and found its exposure to be 20% of plasma exposure, so a plasma half-life stands here as a surrogate for what the tumour sees), the regimens of RTOG 0525 [8] (200 mg/m² on days 1-5 or 100 mg/m² on days 1-21 of 28-day cycles, six cycles), MGMT-expressing fractions of 1% and 30% for methylated and unmethylated promoters (illustrative values bracketing low and high expression rather than measured fractions; methylation shifts graded expression rather than a bimodal positive fraction), and MGMT either stable or consumed stoichiometrically by the drug, one molecule per lesion repaired ($k = 300$, $K_m = 150$ in the expression above). The rate constant was chosen so that the consumption rate at the standard bolus peak equals that of a dose-scaled first-order degradation, which isolates the consequence of letting depletion follow the cumulative exposure instead of the peak concentration; this is the suicide-enzyme behaviour that motivated dose-dense schedules.

Statistics and reproducibility

No statistical method was used to predetermine sample size, and no experimental data were generated in this study; sample sizes are the numbers of simulated cells, founders or seeds stated in each figure legend and in Supplementary Table 2. No data were excluded. Randomisation and blinding do not apply: every result is a deterministic function of an explicit random seed, and all seeds are fixed in the scripts that generate the figures. Replicates are independent simulations differing only in seed; where a mean and a standard deviation are reported, the number of seeds is given in the legend. Kolmogorov-Smirnov distances are compared with the 99% critical value for the stated sample size. Distributional agreement between kernels and with an independent implementation was assessed by two-sample Kolmogorov-Smirnov statistics. Correlations are Pearson correlations unless stated otherwise; correlations of binary fates between related cells are phi coefficients. Posterior summaries are medians, because the posteriors are asymmetric on the log scale. Schedules within a scan are evaluated on common random numbers, the same seeds for every point of the grid, so they are compared by the paired differences over those seeds rather than by an unpaired standard error, which the shared founder population would make invalid; a margin is called resolved when a two-sided paired Student's t-test rejects equality at the 5% level, which on four degrees of freedom requires the margin to exceed 2.776 of its paired standard errors rather than two. Multithreaded runs give results identical to single-threaded runs, because each cell carries its own random number generator seeded from its parent.

Use of artificial intelligence

The software and the first draft of this Article were produced with Claude (Anthropic), a large language model, under the direction and supervision of the author. Specifically, Claude wrote the Julia source of the package, the simulation and plotting scripts, the test suite and the first draft of the text, and reviewed and revised successive drafts, working from the author's specification of the scope, the models, the validation strategy and the case studies. The author defined every scientific question addressed here, chose the models, parameters and reference data, independently verified all numerical results against the generated output tables, checked every

claim against its cited source, and wrote the final version of the manuscript; the author takes full responsibility for the content. The tools used were Claude Opus 5, Claude Opus 4.8 and Claude Fable 5.1 (Anthropic), accessed through the Claude Code command-line interface between 19 and 22 September 2026; the model used for each commit is recorded in that commit’s trailer in the repository, so the record is auditable rather than summary. The model was also used to suggest candidate literature, and most of the references cited here were first surfaced during model-assisted drafting and review. None was cited on the model’s assertion alone: every reference was independently retrieved by the author from the primary record through PubMed, PubMed Central or bioRxiv, and its bibliographic details, together with every quantitative claim drawn from it, were checked against that record before it entered the bibliography. The model is not an author, in keeping with journal policy on generative artificial intelligence, and the author is accountable for the accuracy and integrity of everything reported here.

No display item in this Article is generated by a generative model. Every figure and table is produced deterministically from simulation output by the scripts in the `paper/` directory: the panels by matplotlib from the comma-separated tables in `paper/output/`, and every number quoted in the Results and the Supplementary Information by `paper/manuscript/fill_numbers.py`, which substitutes values read from those tables into the manuscript source. Figure 1 is a schematic drawn by `paper/fig1_schematic.py`. Re-running the scripts from the fixed seeds reproduces all of them.

Data availability

The laboratory and clinical reference values used for calibration and validation were transcribed from the published articles cited in Supplementary Table 4 and are provided, with their provenance, in the `paper/data/` directory of the software repository (<https://github.com/ayoublasri/Biomodelling.jl>). All other data in this Article are simulated: they are regenerated exactly, from fixed seeds, by the scripts in the `paper/` directory of that repository, so those scripts rather than a deposited table are the primary record. The archived release carries them (version DOI 10.5281/zenodo.22894286 for v2.0.0; concept DOI 10.5281/zenodo.22894285).

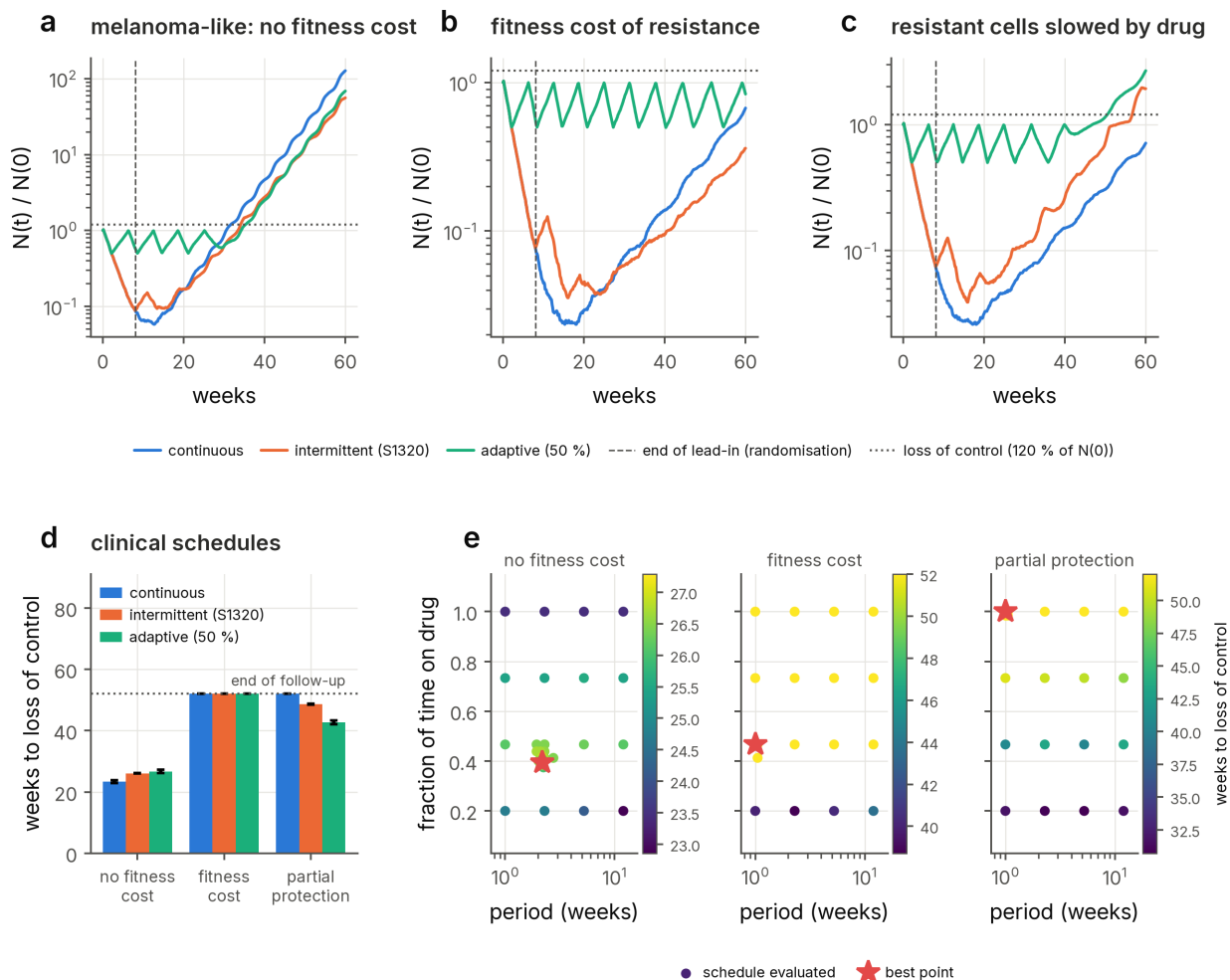
Code availability

Biomodelling.jl 2.0 is open source under the MIT licence at <https://github.com/ayoublasri/Biomodelling.jl>, with documentation, tutorials and a test suite that runs under continuous integration. Every figure is regenerated from fixed seeds by the scripts in `paper/`. After instantiating the environment (`julia --project=paper -e 'using Pkg; Pkg.develop(path="."); Pkg.instantiate()'`), the simulation data are produced by `julia --project=paper -t 4 paper/run_all.jl`, the Python network-inference and imputation benchmarks of the companion Article by `python paper/scripts/benchmarks_python.py`, and the identifiability analysis of the companion Article by `julia --project=paper paper/scripts/fig7_calibration.jl profile`. The remaining opt-in analyses, the cell-cycle refits and scans (`fig7_calibration.jl cycle`, `fig4_persisters.jl cycle seeds`, `fig8_schedules.jl melanoma_cycle melanoma_memory`) and the exact-solution validation of the companion Article (`fig9_validation.jl`, followed by `python paper/scripts/exact_solutions.py`), are listed in `paper/README.md` with their invocations. The panels are then assembled by `python paper/fig1_schematic.py`, `python paper/plot_all.py` and `python paper/plot_supplement.py`. Full instructions, including the four-thread invocation that produces the timings of Supplementary Table 3, are in `paper/README.md`. Version 2.0.0, the version used for every result in this Article, is archived at Zenodo under the version DOI 10.5281/zenodo.22894286, which is frozen to that release; the concept DOI 10.5281/zenodo.22894285 resolves to the latest release instead.

447 Software

448 Biomodelling.jl 2.0 is written in Julia (≥ 1.10) and depends on Distributions.jl, PoissonRandom.jl, QuadGK.jl,
 449 Optim.jl, SpecialFunctions.jl and StatsBase.jl; HDF5.jl is an optional extension. Benchmarks used absolute
 450 Pearson and Spearman correlations and scikit-learn random forests re-implementing GENIE3 [29], MAGIC [30]
 451 and kNN-smoothing [31]. The test suite (Aqua quality checks and statistical tests) runs under GitHub Actions on
 452 Julia 1.10 and the latest release.

453 Figure legends



454

455 **Fig. 1 | Dosing schedules for a melanoma-like population with a heritable pre-resistant state.** a–c,
 456 Population size, relative to its pre-treatment value, under continuous dosing, the intermittent schedule of the
 457 SWOG S1320 trial (8-week lead-in, dashed line, then 3 weeks off and 5 weeks on) and adaptive therapy (drug
 458 withdrawn below 50% of the initial size, resumed at the initial size) for a melanoma-like population with a rare,
 459 heritable pre-resistant state, when the resistant state carries no fitness cost (**a**), when resistant cells grow 50%
 460 slower in the absence of drug (**b**), and when resistant cells are only partly protected and grow at half speed under
 461 drug (**c**); dotted line, 120% of the initial size. **d**, Weeks after randomisation until the population exceeds 120%
 462 of its pre-treatment size (loss of control; mean $\pm$ s.d. of four seeds, censored at 52 weeks). **e**, The same end
 463 point over intermittent schedules of given period and fraction of time on drug after the lead-in, for every schedule

evaluated by the optimiser (star, best point; where several schedules reach the 52-week censoring limit it marks the first of a tie), for the three mechanisms.

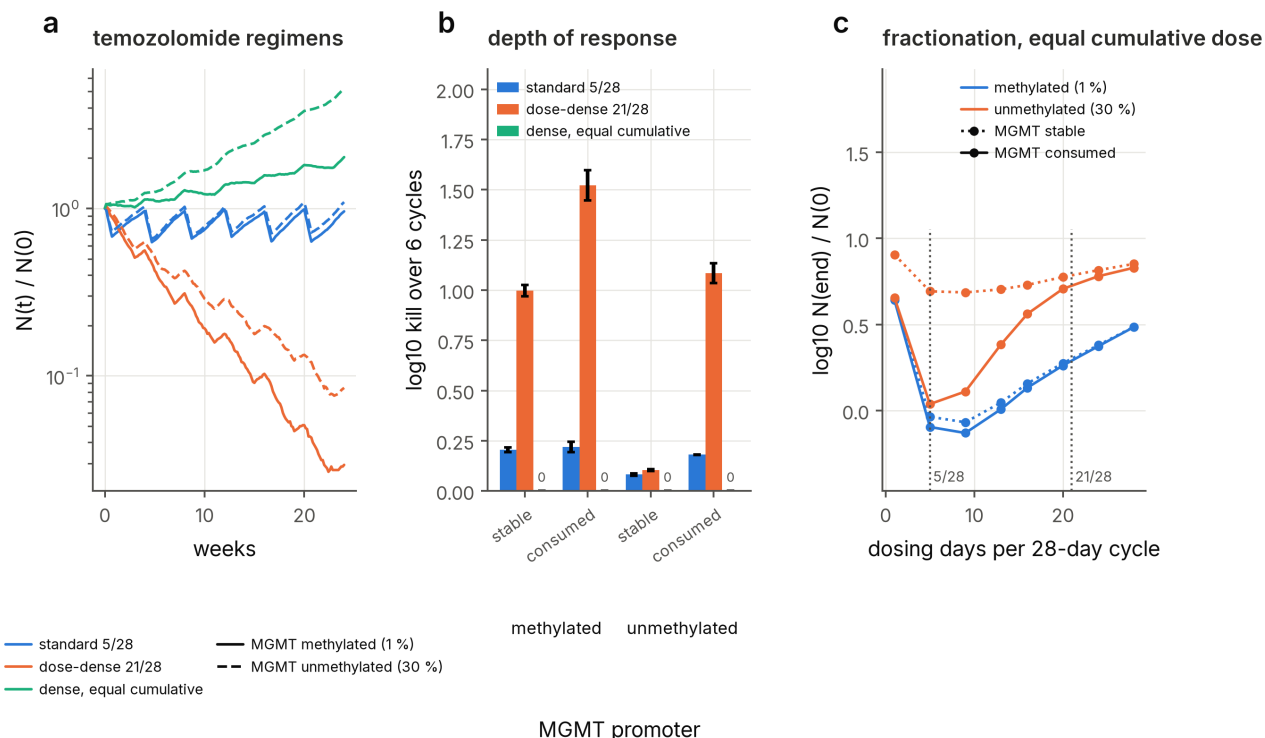


Fig. 2 | Temozolomide regimens of the RTOG 0525 trial with MGMT stable or consumed by the drug. **a**, Population size under the standard (5 of 28 days) and dose-dense (21 of 28 days) temozolomide regimens of the RTOG 0525 trial and a 21-day regimen at the standard cumulative dose, with one-compartment pharmacokinetics (elimination half-life 2.1 h, a plasma value; an independent study gives 1.8 h) and stoichiometric consumption of MGMT by the drug, one molecule per lesion repaired, for populations in which 1% (methylated promoter, solid) or 30% (unmethylated, dashed) of cells express MGMT. **b**, Depth of response over six cycles for the three regimens, the two populations and MGMT either stable or consumed by the drug (mean $\pm$ s.d. of two seeds); a bar at zero marks a regimen that achieves no net kill (the 21-day schedule at the standard cumulative dose). **c**, Final population size against the number of dosing days per 28-day cycle at the standard cumulative dose.

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Version 1 of the framework was developed with Marc Sturrock and Vahid Shahrezaei, whose contributions to that work are described in ref. [32]. This Article and version 2.0 of the software were authored with Claude (Anthropic), a large language model, under the direction and supervision of the author; what the model did and what the author verified independently are set out in the Methods, under Use of artificial intelligence.

Author contributions

A.L. conceived the study, defined its scope, the validation strategy and the case studies, supervised the work, verified all numerical results and biological interpretations against their sources, and wrote the final version of the

manuscript. The implementation of the software, the simulations, the analyses, the figures and the first draft of the manuscript were produced with Claude (Anthropic) under the author’s direction, as described in the Methods. Claude is not an author and no display item was produced by a generative model.

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Competing interests

A.L. is a director of Berache Limited, Dublin, Ireland, a consulting company active in modelling, computational biology and pharmaceutical manufacturing; this commercial activity overlaps the subject area of this Article. A.L. derives no revenue from Biomodelling.jl, which is released under the MIT licence. This does not alter the author’s adherence to the journal’s policies on sharing data and materials.

Additional information

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Supplementary information accompanies this Article. The schedule optimiser and the clinical case studies built on the same framework are reported in a companion Article. Correspondence and requests for materials should be addressed to A.L. (alasri@berache.com).

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